

Signals & Systems (EE504)
Pre-Reading Material – Lecture 2
Signals & Their Transformation
Estimated Reading Time: 20–25 minutes

Course Context

This lecture introduces the fundamental operations performed on signals and the characteristics used to classify them. Students will learn how signals can be transformed through time shifting, scaling, and folding, and how these transformations help engineers analyze real-world systems. The lecture also introduces elementary signals such as exponential, sinusoidal, unit step, and unit impulse signals, which serve as the basic building blocks in signal analysis.

Why This Topic Matters

Signal transformations are widely used in engineering applications such as audio processing, communication systems, image processing, biomedical engineering, and control systems.

For example:

- Playing an audio recording in reverse represents **signal folding**.
- Playing an audio recording faster or slower represents **time scaling**.
- Delaying an audio or communication signal represents **time shifting**.

Understanding signal properties such as **symmetry** and **periodicity** enables engineers to simplify the analysis of complex systems and forms the basis of advanced techniques such as Fourier analysis.

Learning Outcomes

After completing this lecture, students should be able to:

- Explain the basic signal transformations: folding, scaling, and shifting.
- Distinguish between even and odd signals.
- Differentiate between continuous-time and discrete-time periodic signals.

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Course Repository: <https://github.com/rajesh101191/Signals-Systems-PIEMR-2026-EE504>



- Describe the characteristics of exponential and sinusoidal signals.
- Explain the significance of unit step and unit impulse signals in system analysis.

Activate Your Prior Knowledge

In Lecture 1, you learned that a **signal** is a quantity that varies with time (or another independent variable) and carries information, while a **system** processes an input signal to produce an output signal.

Recall that:

- Continuous-time signals exist at every instant of time.
- Discrete-time signals exist only at specific sampling instants.
- Digital systems process discrete-time signals because they provide greater accuracy, reliability, and flexibility.

Prerequisite Concepts

Independent Variable

The variable with respect to which a signal changes, usually time in continuous-time systems or sample index in discrete-time systems.

Uniform Sampling

The process of sampling a continuous-time signal at equal time intervals to obtain a discrete-time signal.

Analysis

Determining the output of a known system for a given input signal.

Synthesis (Design)

Designing a system that produces a desired output for a specified input. Unlike analysis, a synthesis problem may not always have a feasible solution.



Core Concepts

Signal Transformations

Signals can be modified in several ways without changing their fundamental nature.

- **Folding (Time Reversal):** Reflects a signal about the vertical axis. A practical example is playing an audio recording backward.
- **Scaling:** Changes the speed of a signal. Compression makes a signal occur faster, while stretching makes it occur more slowly.
- **Shifting:** Moves a signal along the time axis. A delay shifts the signal to the right, whereas an advance shifts it to the left.

Even and Odd Signals

Signals can be classified according to their symmetry.

- **Even signals** are symmetric about the vertical axis.
- **Odd signals** exhibit rotational symmetry about the origin.

Any signal can be represented as the sum of an even component and an odd component, which simplifies signal analysis.

Periodic Signals

A **periodic signal** repeats itself after a fixed interval.

- A **continuous-time periodic signal** repeats after a constant time interval.
- A **discrete-time periodic signal** repeats after a fixed number of samples. Because the repetition occurs over integer sample indices, the periodicity condition is more restrictive than in continuous time.

Exponential and Sinusoidal Signals

Exponential and sinusoidal signals are among the most important signals in engineering.

- **Growing exponentials** are commonly used to model population growth and similar increasing phenomena.
- **Decaying exponentials** describe processes such as capacitor discharge, radioactive decay, and damping.



- **Sinusoidal signals** are fundamental in communication systems, alternating-current circuits, and signal processing.

Unit Step and Unit Impulse Signals

The **unit step signal** represents a signal that switches from one state to another at a particular instant.

The **unit impulse signal** is an idealized signal that exists only at a single instant and is used extensively to characterize the behavior of systems.

These two signals share an important mathematical relationship and form the foundation for the analysis of linear time-invariant (LTI) systems.

7. Guided Reading Questions

1. How does playing an audio recording backward relate to signal folding?
2. What is the difference between signal compression and signal stretching?
3. Why is the periodicity condition more restrictive for discrete-time signals?
4. Why are exponential signals important in engineering applications?
5. What roles do the unit step and unit impulse signals play in system analysis?

Reflection Activity

Think of a real-world application involving signal transformations.

Examples include:

- Audio editing software
- Video playback
- Mobile communication systems
- Medical signal processing (ECG or EEG)

Identify:

- the input signal,
- the transformation performed,

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- and the resulting output signal.

How does the transformation improve or modify the information contained in the original signal?

Self-Assessment

True or False

An odd signal can have a non-zero value at the origin.

Answer: False.

Short Answer

What is the difference between signal compression and signal stretching?

Answer: Signal compression reduces the time duration of a signal, making it occur faster, whereas signal stretching increases its time duration, making it occur more slowly.

Think About It

Why are unit step and unit impulse signals considered the fundamental building blocks for analyzing engineering systems?

